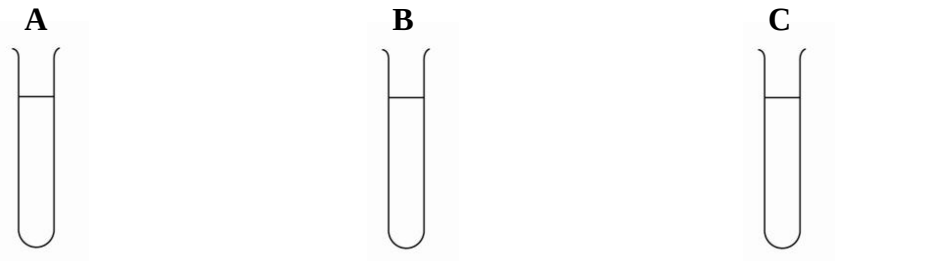
## Short questions

|!|BCESB0070000001|!| ★ **SI**

Anne carried out an investigation to study the digestion of lipid by the enzyme lipase. She added peanut oil and lipase to a test tube and then immediately measured the pH of the mixture. After 15 minutes, she measured the pH of the mixture again.

(a) Anne found that the pH of the mixture changed after 15 minutes. Describe and explain the change in pH. (2 marks)

(b) Anne set three boiling tubes as shown below.




| A | B | C |
|---|---|---|
| oil + lipase + bile salts + pH indicator | oil + lipase + distilled water + pH indicator | oil + boiled lipase + bile salts + pH indicator |

She measured the length of time it took for the indicator to change colour and recorded the results in the table below.

| **Tube** | **Time taken for the indicator to change colour (minute)** |
|---|---|
| A | 2 |
| B | 6 |
| C | No change |

Explain the results of the three tubes. (5 marks)

© 2020 Aristo Educational Press Ltd.

##

(a) Lipase catalyses the breakdown of lipids to glycerol and fatty acids. (1)
Fatty acids lower the pH of the mixture. (1)

(b) In tube A, bile salts emulsified lipids into small droplets. (1)
This increased the surface area of the lipids for the action of lipase. (1)
The rate of lipid digestion was high, (1)
so it took a short time for the indicator to change colour.
In tube B, the rate of lipid digestion in the absence of bile salts was low, (1)
so it took longer time for the indicator to change colour.
In tube C, lipase had been denatured by boiling, so no digestion of lipid took place. (1)

(7 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Applying concepts | 2 |
| (b) | Applying concepts | 5 |

|!|BCESA0070000002|!| ★★ **STSE**

Acute pancreatitis may occur when gallstones block the pancreatic duct. The pancreas becomes inflamed, causing upper abdominal pain, fever and vomiting. Blood tests that indicate the presence of lipase and amylase suggest the occurrence of acute pancreatitis. Treatment may require surgery to remove the gallstones.

(a) Suggest the effects of a blocked pancreatic duct on digestion. (4 marks)

(b) If the pancreatic duct remains blocked for too long without treatment, pancreatic juice may damage the pancreas. Explain how this may happen. (3 marks)

© 2020 Aristo Educational Press Ltd.

##

(a) Pancreatic juice which contains amylase and lipase cannot reach the duodenum. (1)
Starch in food would be partially digested by salivary amylase only. Without pancreatic amylase, not all the starch can be digested. (1)
Without lipase, lipids would not be digested. (1)
Lipids would be passed out of the body in faeces. (1)

(b) Pancreatic juice which contains lipase and protease may accumulate around the pancreas. (1)
Lipase catalyses the breakdown of lipids in the cell membrane. (1)
Protease catalyses the breakdown of protein structures in cells. (1)

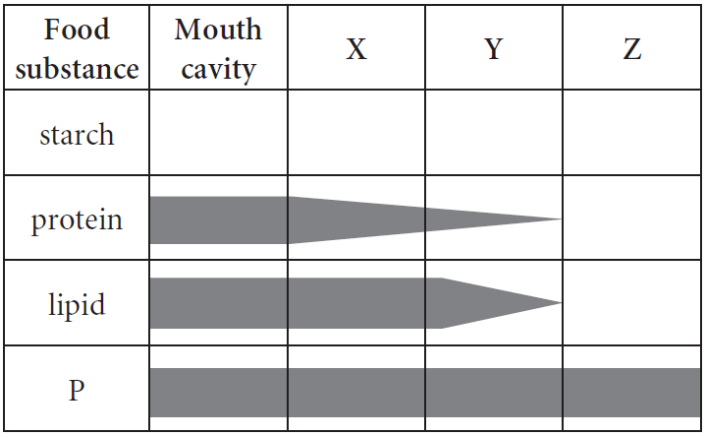
(7 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Applying concepts | 4 |
| (b) | Applying concepts | 3 |

|!|BCESA0070000003|!| ★★

The diagram below shows the digestion of different types of food substances as they pass along the alimentary canal. The width of each band indicates the concentration of each type of food substances.




(a) Identify regions X and Y. (2 marks)

(b) Explain why the concentration of lipid does not decrease immediately after entering region Y. (2 marks)

(c) What is food substance P? Give a reason to support your answer. (2 marks)

(d) Complete the diagram to show the digestion of starch as it passes along the alimentary canal. (1 mark)

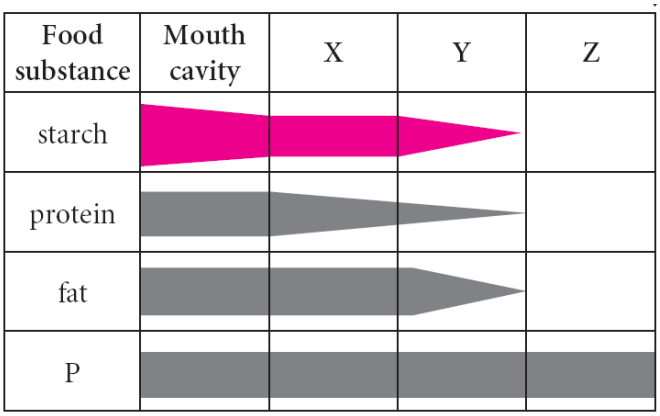
##

(a) X is the stomach. (1)
Y is the small intestine. (1)

(b) In region Y, lipid is mixed with bile which causes the emulsification of lipid. (1)
Lipase (from the pancreatic juice) then acts on the small fat droplets. (1)

(c) P is dietary fibre. (1)
The human digestive system does not produce suitable enzymes to digest dietary fibre. (1)

(d)




(1)

(7 marks)

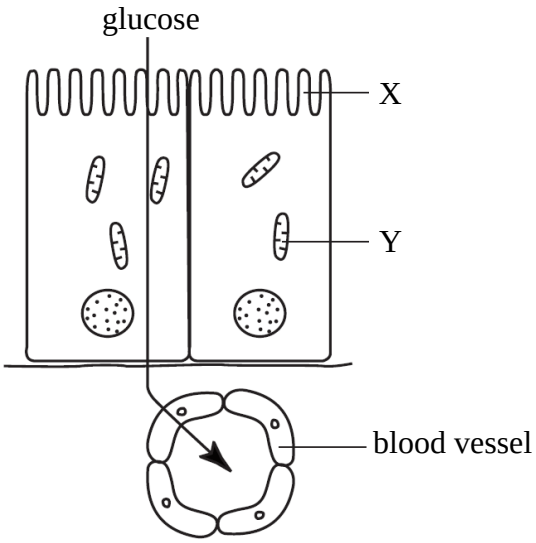
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 2 |
| (b) | Applying concepts | 2 |
| (c) | Analysing data | 1 |
| | Applying concepts | 1 |
| (d) | Applying concepts | 1 |

© 2020 Aristo Educational Press Ltd.

|!|BCESB0070000004|!| ★

The diagram below shows two epithelial cells of a villus and the movement of glucose into a blood vessel.




(a) Name structure X and organelle Y. (2 marks)

(b) Describe how X and Y assist in the absorption of glucose. (3 marks)

(c) What will happen to the absorbed glucose? (2 marks)

##

(a) X is a microvillus. (1)
Y is a mitochondrion. (1)

(b) X can increase the surface area for absorption of glucose. (1)
Y can carry out respiration to release energy. (1)
Glucose is absorbed by active transport which requires energy. (1)

(c) Absorbed glucose will be broken down to release energy in body cells (1)
or converted into glycogen in the liver for storage. (1)

(7 marks)

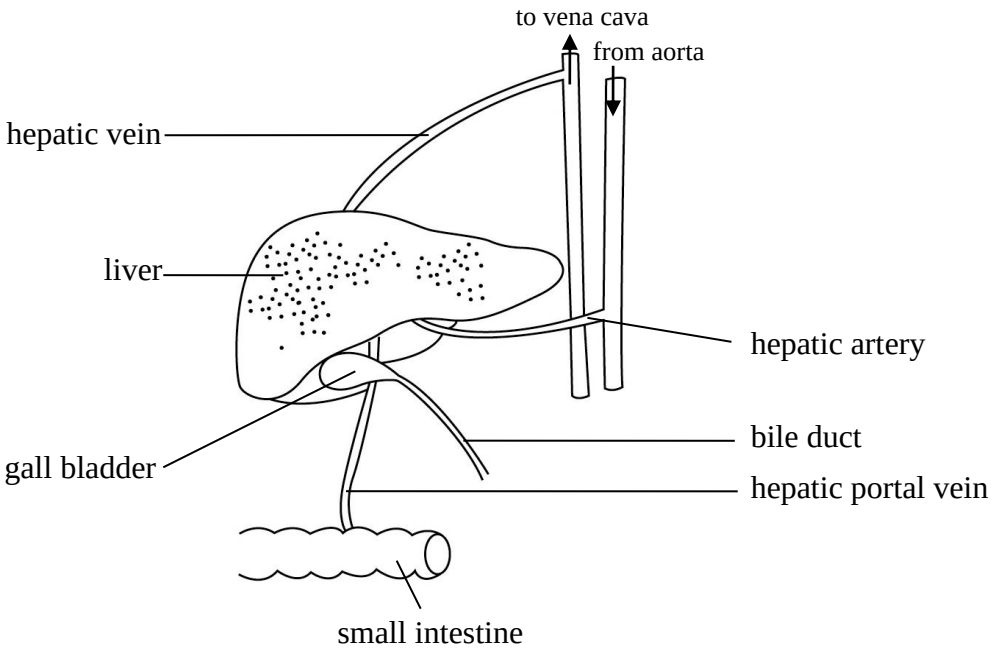
© 2020 Aristo Educational Press Ltd.

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 2 |
| (b) | Applying concepts | 3 |
| (c) | Understanding basic concepts | 2 |

|!|BCESC0070000005|!|

The diagram below shows the liver and its blood supply.




(a) Besides digestive function, state ***two*** other functions of the liver. (2 marks)

(b) Name the blood vessel

(i) with the highest glucose content after a meal. (1 mark)

(ii) with the highest urea content. (1 mark)

(iii) with the highest glucose content during prolonged starvation. (1 mark)

© 2020 Aristo Educational Press Ltd.

##

(a) Break down excess amino acids into urea /
Help to maintain a stable blood glucose level by converting glucose to glycogen or glycogen to glucose /
Storage of iron and fat-soluble vitamins /
Detoxification / break down toxic substances /
Destruction of red blood cells / break down haemoglobin into bile pigment
(Any ***two***, 2)

(b) (i) hepatic portal vein (1)
(ii) hepatic vein (1)
(iii) hepatic vein (1)

(5 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Understanding basic concepts | 2 |
| (b)(i) | Applying concepts | 1 |
| (b)(ii) | Applying concepts | 1 |
| (b)(iii) | Applying concepts | 1 |

© 2020 Aristo Educational Press Ltd.

|!|BCESA0070000006|!| ★★ **SI**

A group of students investigated the action of lipase on the digestion of lipids in milk. They set up three test tubes A, B and C which were incubated at 37 °C. The following table shows the contents of the test tubes.

| Test tube | Contents |
|---|---|
| A | milk + lipase |
| B | milk + lipase + bile salts |
| C | milk + bile salts |

The following table shows the results.

| Time (minutes) | pH | | |
|---|---|---|---|
| | Tube A | Tube B | Tube C |
| 0 | 8.5 | 8.5 | 8.5 |
| 10 | 8.0 | 7.7 | 8.5 |
| 20 | 7.5 | 7.0 | 8.5 |
| 30 | 7.3 | 6.5 | 8.5 |
| 40 | 7.0 | 6.5 | 8.5 |
| 50 | 6.5 | 6.5 | 8.5 |
| 60 | 6.5 | 6.5 | 8.5 |

(a) Explain the pH change of the solution in test tube A. (2 marks)

(b) Explain why the pH value of the solution in tube A did not fall below 6.5. (1 mark)

(c) The rate at which the pH fell in tube A was different from that in tube B. Explain the difference. (2 marks)

(d) Explain why test tube C was set up. (1 mark)

##

© 2020 Aristo Educational Press Ltd.

(a) Lipase catalyses the digestion of lipids into glycerol and fatty acids. (1)
Fatty acids lower the pH of the solution. (1)
(b) The substrate (lipid) has been used up. /
The lipase has become denatured at pH 6.5. (1)
(c) Bile salts emulsify lipids into small droplets in tube B. (1)
This increases the surface area of lipids for lipase to act on. (1)
Therefore, the rate of digestion of lipids in tube B is faster.
(d) to show that lipase has to be present for the pH to change (1)

(6 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 2 |
| (b) | Applying concepts | 1 |
| (c) | Applying concepts | 2 |
| (d) | Practical skills | 1 |

|!|BCESC0070000007|!|
Fill in the blanks to complete the following paragraph. (6 marks)

Digestion of food starts in the mouth. The food is chewed by the teeth, mixed with (a)_____ and rolled into a bolus, which is swallowed down the (b)_____ and then enters the (c)_____. The chyme formed then passes into the duodenum, which receives (d)_____ released from the gall bladder and the (e)_____ released from the pancreas. The digestion of carbohydrates, lipids and proteins is finally completed in the (f)_____.

##

(a) saliva (1)
(b) oesophagus (1)
(c) stomach (1)
(d) bile (1)
(e) pancreatic juice (1)
(f) small intestine (1)

(6 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Understanding basic concepts | 1 |
| (b) | Understanding basic concepts | 1 |

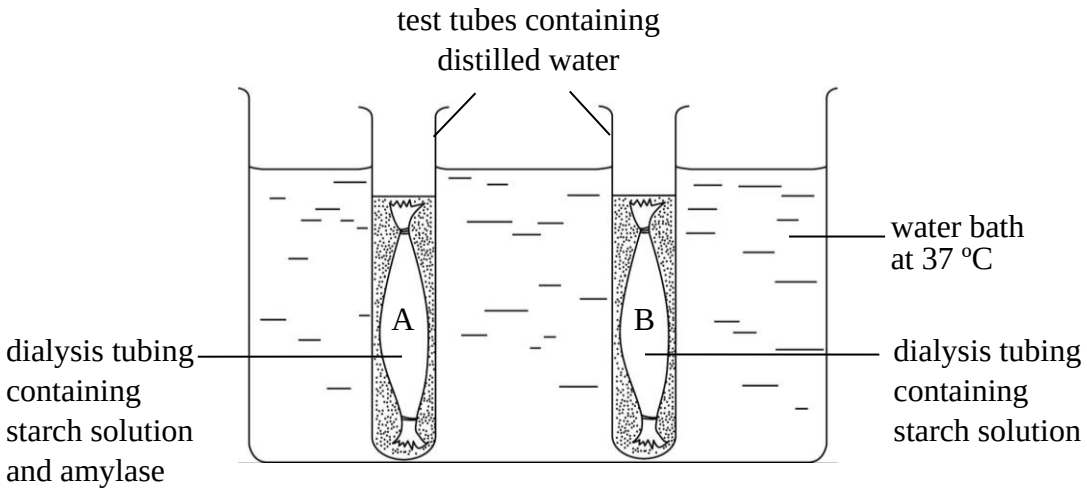
© 2020 Aristo Educational Press Ltd.

| | | |
|---|---|---|
| (c) | Understanding basic concepts | 1 |
| (d) | Understanding basic concepts | 1 |
| (e) | Understanding basic concepts | 1 |
| (f) | Understanding basic concepts | 1 |

|!|BCESB0070000008|!| ★ **SI**

The diagram below shows an experimental set-up. The dialysis tubings were immersed in distilled water in two test tubes which were kept in a 37 °C water bath.

At the beginning of the experiment, the water in the test tubes was tested for starch and reducing sugars. It was found that neither starch nor reducing sugar was present in the water in the test tubes. After one hour, the water in the test tubes was tested for starch and reducing sugars again.




(a) Suggest a test for starch and reducing sugars respectively. (2 marks)

(b) What would be the expected test results after one hour? (2 marks)

(c) If the experiment were repeated with the two test tubes placed in a refrigerator instead of a 37 °C water bath, how would you expect the results to differ? (1 mark)

© 2020 Aristo Educational Press Ltd.

(d) The set-up can represent a part of the digestive system. Which part of the human body does the liquid surrounding the dialysis tubing represent? (1 mark)

---

##

(a) Test for starch: iodine test (1)

Test for reducing sugars: Benedict's test (1)

(b)

| | **Iodine test** | **Benedict's test** | |
|---|---|---|---|
| Dialysis tubing A | Negative results: the iodine solution remains brown. | Positive results: brick-red precipitates are formed. | (1) |
| Dialysis tubing B | Negative results: the iodine solution remains brown. | Negative results: the mixture remains clear blue. | (1) |

(c) Reducing sugars would be absent in the water surrounding dialysis tubing A. (1)

(d) blood (1)

(6 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Understanding basic concepts | 2 |
| (b) | Analysing data | 2 |
| (c) | Applying concepts | 1 |
| (d) | Analysing data | 1 |

|!|BCESA0070000009|!| ★★ **STSE**

People with lactose intolerance lack the enzyme lactase in their small intestine. They have a high chance of diarrheoa after drinking milk.

(a) Using your knowledge about the digestion of carbohydrates in the human body, explain why a person with lactose intolerance has diarrheoa after drinking milk. (4 marks)

---

---

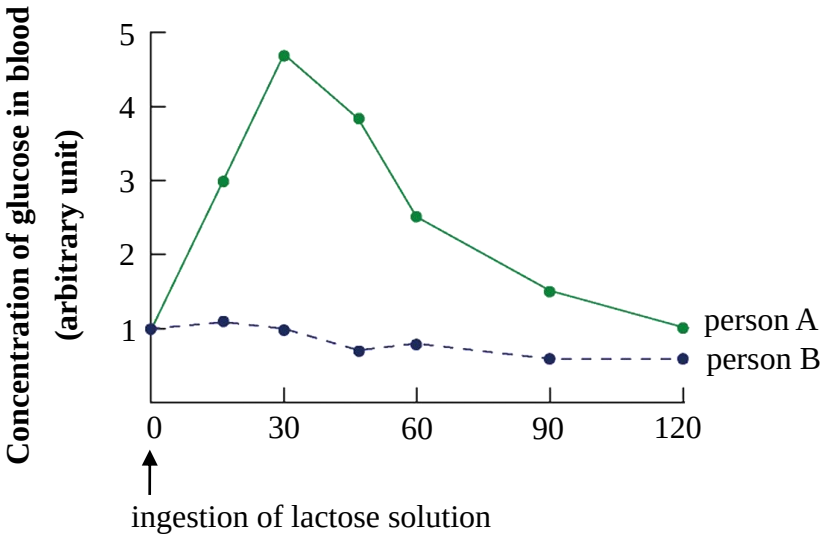
---

---

---

© 2020 Aristo Educational Press Ltd.

(b) A doctor tested two people, A and B, for lactose intolerance. Both people were fasted for 10 hours. They were then given a lactose solution to drink. The concentration of glucose in their blood was measured over the next 2 hours. The results are shown in the graph below.




Deduce from the graph, which person has lactose intolerance. (3 marks)

##

(a) Lactose is a sugar found in milk. (1)
The patient lacks lactase, so lactose cannot be digested and it remains in the intestines. (1)
The water potential of the content inside the intestines becomes lower. (1)
As a result, less water is absorbed by osmosis (1)
and watery faeces will form.

(b) Person B has lactose intolerance. (1)
Person B lacks lactase and cannot digest lactose into glucose and galactose. (1)
No glucose was formed or absorbed into his/her blood, so his/her blood glucose level remained unchanged after the ingestion of lactose solution. (1)

(7 marks)

© 2020 Aristo Educational Press Ltd.

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Applying concepts | 4 |
| (b) | Analysing data | 1 |
| | Applying concepts | 2 |

|!|BCESC0070000010|!|

For each of the organs listed in ***Column 1***, select from ***Column 2*** one function in nutrition that matches it. Put the appropriate letter in the space provided. (3 marks)

| ***Column 1*** | | ***Column 2*** |
|---|---|---|
| Liver | (a) _____ | A. site of protein digestion |
| Pancreas | (b) _____ | B. site of bile production |
| Stomach | (c) _____ | C. site of lipid digestion |
| | | D. site of absorption of amino acid |
| | | E. site of production of hormones that regulate the blood glucose level |

##

(a) B (1)

(b) E (1)

(c) A (1)

(3 marks)

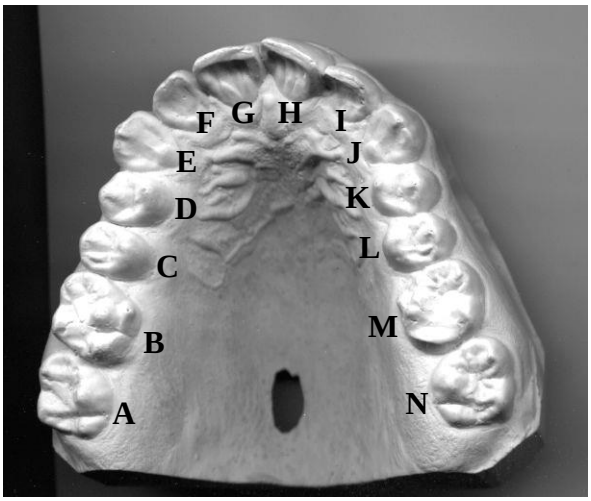
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Understanding basic concepts | 1 |
| (b) | Understanding basic concepts | 1 |
| (c) | Understanding basic concepts | 1 |

© 2020 Aristo Educational Press Ltd.

|!|BCESC0070000011|!|

The photograph below shows the arrangement of four types of teeth in the upper jaw of a man.




(a) Identify which teeth are the incisors and molars, respectively. (2 marks)

Incisors: _____

Molars: _____

(b) Suppose the arrangement of the teeth in the lower jaw is the same, write down the dental formula of this person. (1 mark)

_____

_____

##

(a) Incisors: F, G, H and I (1)

Molars: A, B, M and N (1)

(b)

$$i\ \frac{2}{2}\quad c\ \frac{1}{1}\quad pm\ \frac{2}{2}\quad m\ \frac{2}{2}$$

(1)

(3 marks)

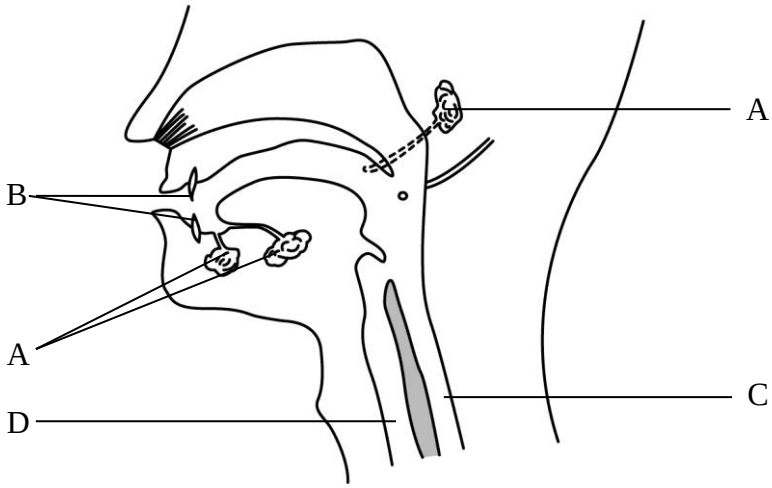
##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Analysing data | 2 |
| (b) | Applying concepts | 1 |

© 2020 Aristo Educational Press Ltd.

|!|BCESC0070000012|!|

The diagram below shows the upper part of the human alimentary canal. Physical and chemical digestion of food occurs in the mouth cavity.




(a) Describe the physical digestion of food by a labelled part in the mouth cavity. (1 mark)

(b) (i) What is the name of the fluid produced by structure A? (1 mark)

(ii) Describe how the fluid produced by structure A is involved in chemical digestion of food in the mouth cavity. (2 marks)

(c) Describe how food is moved along the alimentary canal by peristalsis. (3 marks)

##

(a) The chewing action of B (teeth) mechanically breaks food into smaller pieces. (1)

(b) (i) saliva (1)

(ii) Saliva contains salivary amylase (1)

© 2020 Aristo Educational Press Ltd.

which catalyses the digestion of starch to maltose. (1)

(c) Behind the food, the circular muscles of the alimentary canal contract, pushing the food forward. (1)

The circular muscles in front of food relax. The lumen becomes larger for the food to pass through. (1)

The waves of alternate muscle contractions propel the food forward. (1)

(7 marks)

##

| | Skills assessed | Marks |
|---|---|---|
| (a) | Applying concepts | 1 |
| (b)(i) | Understanding basic concepts | 1 |
| (b)(ii) | Understanding basic concepts | 2 |
| (c) | Understanding basic concepts | 3 |

© 2020 Aristo Educational Press Ltd.